

Experiment no. 07

AIM: Use project management tool to prepare schedule for the project.

Gantt chart, PERT

Theory :

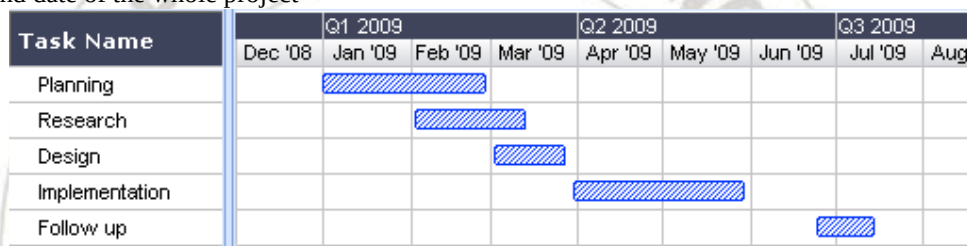
The main aim of PROJECT SCHEDULING AND TRACKING is to get the project completed on time. Program evaluation and review technique (PERT) and Gantt chart are two project scheduling methods that can be applied to software development

Split the project into tasks and estimate time and resources required to complete each task. Organize tasks concurrently to make optimal use of workforce. Minimize task dependencies to avoid delays caused by one task waiting for another to complete

Gantt chart:

A Gantt chart, commonly used in project management, is one of the most popular and useful ways of showing activities (tasks or events) displayed against time. On the left of the chart is a list of the activities and along the top is a suitable time scale. Each activity is represented by a bar; the position and length of the bar reflects the start date, duration and end date of the activity. This allows you to see at a glance:

- What the various activities are
- When each activity begins and ends
- How long each activity is scheduled to last
- Where activities overlap with other activities, and by how much
- The start and end date of the whole project



References:

1. <http://www.gantt.com/>

RMMM plan

The risk components are defined in the following manner:

- **Performance risk**—the degree of uncertainty that the product will meet its requirements and be fit for its intended use.
- **Cost risk**—the degree of uncertainty that the project budget will be maintained.
- **Support risk**—the degree of uncertainty that the resultant software will be easy to correct, adapt, and enhance.
- **Schedule risk**—the degree of uncertainty that the project schedule will be maintained and that the product will be delivered on time.

The impact of each risk driver on the risk component is divided into one of four impact categories—**negligible, marginal, critical, or catastrophic**.

Components		Performance	Support	Cost	Schedule
Category					
Catastrophic	1	Failure to meet the requirement would result in mission failure		Failure results in increased costs and schedule delays with expected values in excess of \$500K	
	2	Significant degradation to nonachievement of technical performance	Nonresponsive or unsupportable software	Significant financial shortages, budget overrun likely	Unachievable IOC
Critical	1	Failure to meet the requirement would degrade system performance to a point where mission success is questionable		Failure results in operational delays and/or increased costs with expected value of \$100K to \$500K	
	2	Some reduction in technical performance	Minor delays in software modifications	Some shortage of financial resources, possible overruns	Possible slippage in IOC
Marginal	1	Failure to meet the requirement would result in degradation of secondary mission		Costs, impacts, and/or recoverable schedule slips with expected value of \$1K to \$100K	
	2	Minimal to small reduction in technical performance	Responsive software support	Sufficient financial resources	Realistic, achievable schedule
Negligible	1	Failure to meet the requirement would create inconvenience or nonoperational impact		Error results in minor cost and/or schedule impact with expected value of less than \$1K	
	2	No reduction in technical performance	Fully supportable software	Possible budget overrun	Early completion

Impact assessment

1) Develop a risk table

Risks	Category	Probability	Impact	RMMM
Size estimate may be significantly low	PS	60%	2	
Larger number of users than planned	PS	30%	3	
Less reuse than planned	PS	70%	2	
End-users resist system	BU	40%	3	
Delivery deadline will be tightened	BU	50%	2	
Funding will be lost	CU	40%	1	
Customer will change requirements	PS	80%	2	
Technology will not meet expectations	TE	30%	1	
Lack of training on tools	DE	80%	3	
Staff inexperienced	ST	30%	2	
Staff turnover will be high	ST	60%	2	
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For risks with high impact and high probability create a RMMM plan

Risk information sheet			
Risk ID: PO2-432	Date: 5/9/02	Prob: 80%	Impact: high
Description: Only 70 percent of the software components scheduled for reuse will, in fact, be integrated into the application. The remaining functionality will have to be custom developed.			
Refinement/context: Subcondition 1: Certain reusable components were developed by a third party with no knowledge of internal design standards. Subcondition 2: The design standard for component interfaces has not been solidified and may not conform to certain existing reusable components. Subcondition 3: Certain reusable components have been implemented in a language that is not supported on the target environment.			
Mitigation/monitoring: 1. Contact third party to determine conformance with design standards. 2. Press for interface standards completion; consider component structure when deciding on interface protocol. 3. Check to determine number of components in subcondition 3 category; check to determine if language support can be acquired.			
Management/contingency plan/trigger: RE computed to be \$20,200. Allocate this amount within project contingency cost. Develop revised schedule assuming that 18 additional components will have to be custom built; allocate staff accordingly. Trigger: Mitigation steps unproductive as of 7/1/02			
Current status: 5/12/02: Mitigation steps initiated.			
Originator: D. Gagne		Assigned: B. Laster	

Task Breakdown:

Phase	Task
Requirement	SRS Preparation
Design	UI + Database Design
Development	Patient Module
Development	Doctor Module
Development	Appointment Module
Development	Billing Module
Development	Login/Auth System
Testing	Unit + Integration Testing
Deployment	System Deployment

Gantt Chart:

Gantt Chart for Hospital Management System (SDLC Timeline)								
Task	W1	W2	W3	W4	W5	W6	W7	W8
SRS Preparation								
System Design								
Database Design								
Patient Module Dev								
Doctor Module Dev								
Appointment Module								
Billing Module Dev								
Authentication Module								
Integration								
Testing								
Deployment								

Dependencies:

- Development depends on Design & DB Design
- Integration depends on all modules
- Testing depends on Integration
- Deployment depends on Testing

Outcomes: Upon completion of the course students will be able to draw the gantt chart for the project.

Conclusion:

For Faculty Use

Correction Parameters	Formative Assessment [40%]	Timely completion of Practical [40%]	Attendance / Learning Attitude [20%]	
Marks Obtained				